

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1	(a)	(i)		Aberystwyth has less radon than Penzance but more than Chester (1) OWTTE and the pie chart for location A has less radon than location C but more than location B (1) [so Adam is correct]			2	2		
		(ii)	I	$\frac{30}{60} = 0.5$ [cps]		1		1	1	1
			II	15		1		1	1	1
		(iii)		Man-made sources are only 14.3% or 14.5% (1) this is lower than natural sources (1) ORA so disagree Conclusion must be present to award both marks OR Radon [is a natural source and it] is 50% (1) So man-made sources can't be more than 50% (1) so disagree Conclusion must be present to award both marks OR Correctly named sources of natural radiation are stated with percentage stated (1) So there is a larger proportion from natural sources (1) so disagree Conclusion must be present to award both marks			2	2		
	(b)	(i)		Attempt at adding her total dose (1) Total dose is $2.7 + 0.18 + 2 \times 0.09 + 0.005 + 6.6 = 9.665$ [units] [this is less than 20] so it is still safe [for her to work with radiation] (1)			2	2	1	